
KESTREL

The drone security analyst that never blinks

Architecture Dossier

Design decisions, data contracts, and the reasoning behind them

Author: Atharva Awade

Companion to the main report

Every figure in this document is generated from a measured run. Re-running the evaluation suite regenerates the tables; no number is transcribed by hand.

FlytBase AI Engineer assignment

Contents

1	Container view	2
2	The frame lifecycle	2
2.1	Why entity resolution must precede rule evaluation	2
2.2	Why baseline observation must follow it	2
3	Memory	2
4	Architecture decision records	4
4.1	ADR 0001 — Model selection by measurement	4
4.2	ADR 0002 — Detection runs on-device	4
4.3	ADR 0003 — SQLite over a vector database	5
4.4	ADR 0004 — Reciprocal Rank Fusion over a learned reranker	5
5	Data contracts	5
5.1	SceneGraph — what tier 3 must return	5
5.2	AlertLocation — what makes an alert dispatchable	5
5.3	The rule language	5
6	Failure modes designed for	6
7	Security boundaries	6

1 Container view

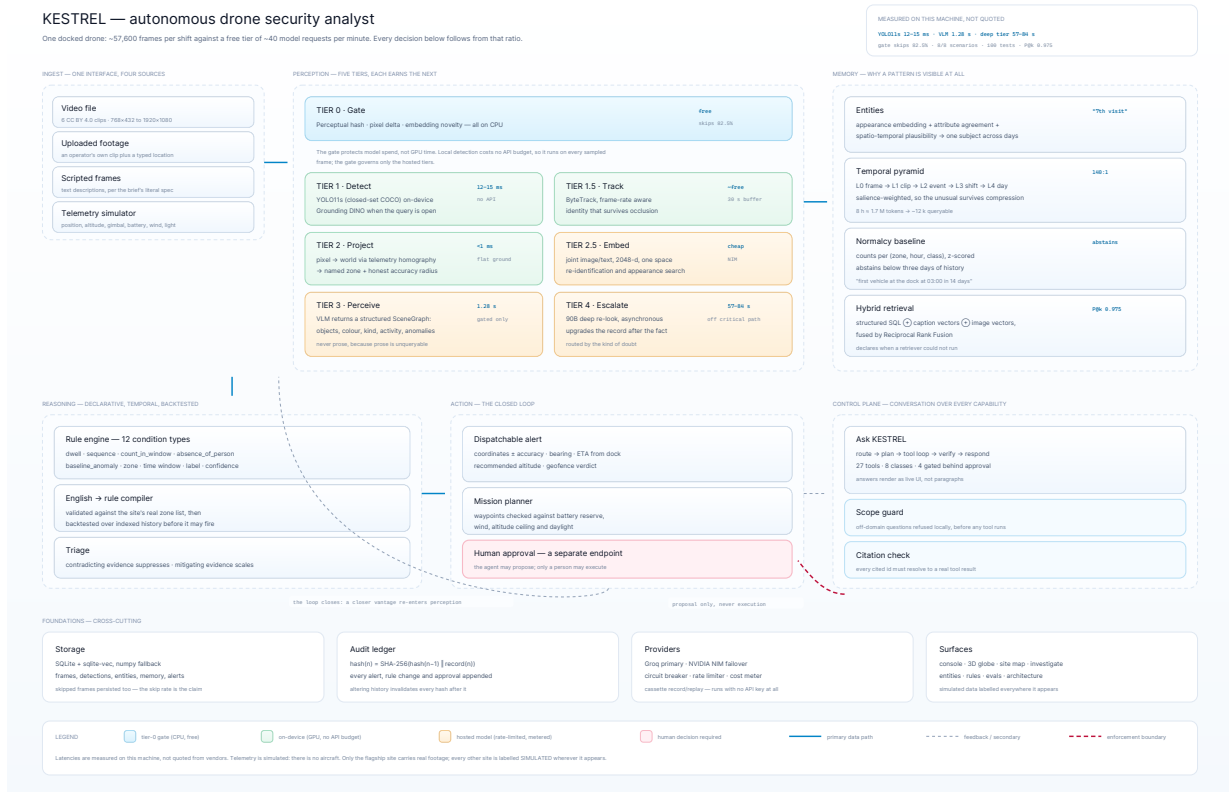


Figure 1: Container architecture. The conversational control plane spans every capability; the cascade runs left to right; the action loop closes back into perception.

2 The frame lifecycle

2.1 Why entity resolution must precede rule evaluation

Dwell is a property of an *entity*, not of a frame. A rule evaluated before identity is assigned has nothing to accumulate against, so loitering can never fire regardless of how long someone stands at the gate. There is no error, no warning, and no symptom other than silence.

2.2 Why baseline observation must follow it

If the frame being judged is folded into the baseline before it is judged, it becomes part of the “normal” it is measured against. Nothing is ever anomalous, and again the only symptom is silence.

3 Memory

An eight-hour shift is roughly 1.7M tokens of raw observation. An operator asking “what happened last night” expects an answer grounded in all of it. Uniform summarisation would discard precisely the material an investigation needs, so compression is weighted by saliency and every node retains its children — a summary that cannot be expanded back to its evidence is not evidence.

4 Architecture decision records

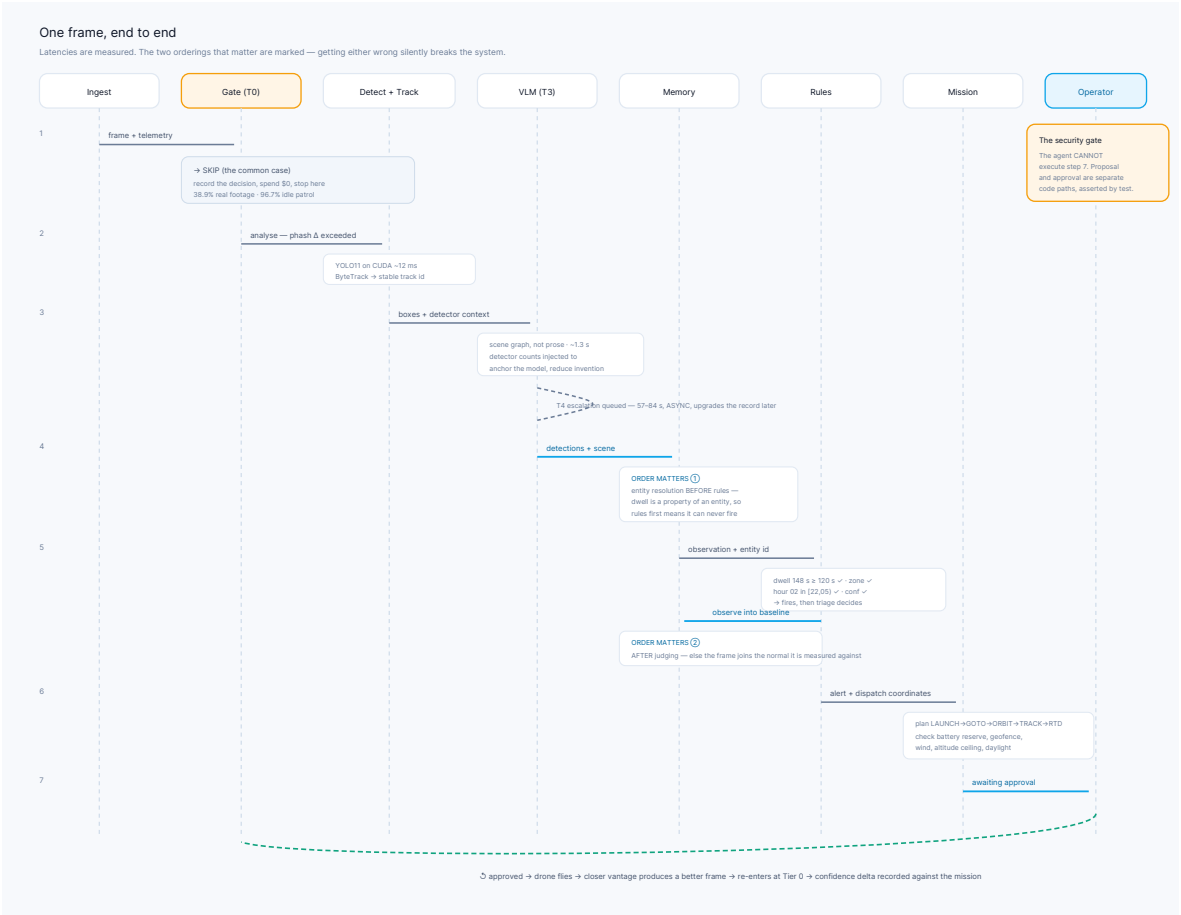


Figure 2: One frame end to end. The two orderings marked in the diagram are the ones that silently break the system when reversed.

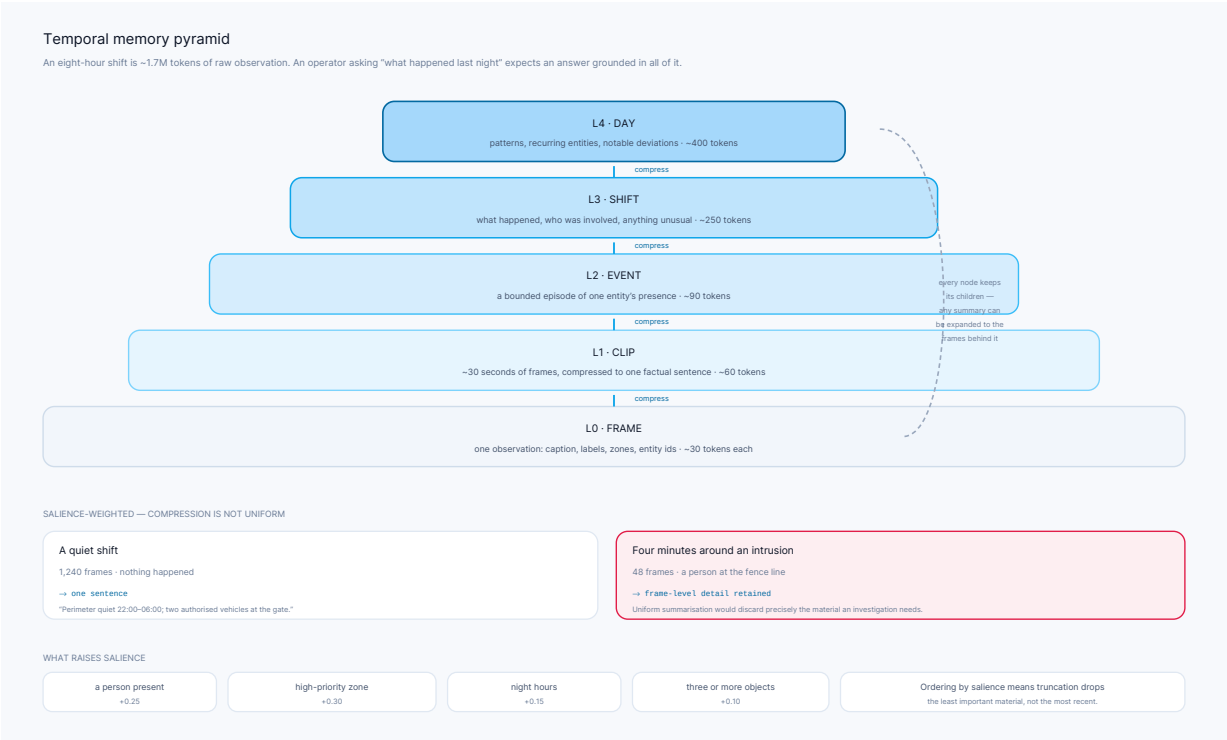


Figure 3: Hierarchical compression. Saliience-weighted, so an uneventful shift becomes a sentence while the minutes around an intrusion keep frame-level detail.

4.1 ADR 0001 — Model selection by measurement

Status: accepted. **Consequence:** four of five original model choices changed, plus one architectural decision (asynchronous deep escalation).

Every model ID was verified against the live API before dependent code was written. Presence in a provider’s catalogue is not evidence that the endpoint is reachable.

Table 1: Endpoint verification. Run before any dependent code was written.

Endpoint	Latency	Status
GET /models	237 ms	PASS
chat nvidia llama-3.1-70b (failover path)	52008 ms	PASS
chat nvidia llama-3.3-70b (unusably slow)	35814 ms	PASS
chat structured json_schema	88589 ms	PASS
vlm llama-3.2-11b-vision	1292 ms	PASS
vlm llama-3.2-90b-vision	90063 ms	n/a
embed nvclip text (unprovisioned, superseded)	110 ms	n/a
embed nvclip image (unprovisioned, superseded)	95 ms	n/a
embed joint text (nemotron-embed-v1)	489 ms	PASS
embed joint image (nemotron-embed-v1)	610 ms	PASS
embed cross-modal alignment	1075 ms	PASS
embed nv-embedqa-e5-v5	373 ms	PASS
rerank llama-3.2-nv-rerankqa	66 ms	n/a
detect nv-grounding-dino	511 ms	n/a
groq llama-3.3-70b-versatile (primary)	271 ms	PASS

4.2 ADR 0002 — Detection runs on-device

Status: accepted.

No hosted detector is reachable. Detection therefore runs locally, which is the correct place for per-frame work regardless: no API budget, no rate limit, and it keeps working with the network unplugged.

The backend is selected by capability at startup and reported honestly in the health endpoint, so the interface can state which one produced a given box:

1. **Grounding DINO** — open-vocabulary, detects from text phrases. Requires the HuggingFace Hub.
2. **YOLO11** — closed-set COCO, weights from GitHub releases. The backend that works when the Hub is blocked, which it was on the build network.
3. **RT-DETR** — closed-set, Apache-2.0.
4. **Motion heuristics** — no torch at all. Degraded but honest: the pipeline still runs, and says that detection is degraded rather than presenting guesses as classifications.

Licence note

Ultralytics YOLO is AGPL-3.0. It is an *optional* extra, never a required dependency, and the primary paths are Apache-2.0. Footage is CC BY 4.0 with attribution recorded in `data/footage/SOURCES.md`. VisDrone and similar academic surveillance datasets were considered and rejected: CC BY-NC-SA permits academic use only, and this submission goes to a company.

4.3 ADR 0003 — SQLite over a vector database

Status: accepted.

A reviewer must be able to clone the repository and run it. A single file with no server, no container and no migration step is worth more here than a distributed index we would never load enough data to justify. If the `sqlite-vec` extension fails to load, a numpy brute-force scan takes over — at this scale a linear scan over a few thousand vectors is milliseconds, and correctness of setup beats asymptotics.

4.4 ADR 0004 — Reciprocal Rank Fusion over a learned reranker

Status: accepted.

No hosted cross-encoder is reachable. RRF requires no model, is deterministic, and combines *rankings* rather than scores — so two retrievers producing incomparable similarity scales can be merged without inventing an arbitrary normalisation. An optional LLM rerank refines the top-*k* where precision matters more than latency.

5 Data contracts**5.1 SceneGraph — what tier 3 must return**

Structured, not prose. A caption is unqueryable.

```
class SceneGraph(BaseModel):
    caption: str
    objects: list[SceneObject]          # label, colour, kind, activity, count, confidence
    activities: list[str]
    lighting: Literal["daylight", "dusk", "night", "artificial", "unknown"]
    visibility: Literal["clear", "reduced", "poor", "unknown"]
    anomalies: list[str]                # empty is the correct answer most of the time
    confidence: float
    tier: Literal["fast", "deep"]
```

5.2 AlertLocation — what makes an alert dispatchable

```
class AlertLocation(BaseModel):
    lat: float | None; lon: float | None
    source: Literal["geo-projection", "zone-centroid", "drone-position", "unknown"]
    accuracy_m: float | None           # provenance travels with the coordinates
    distance_from_dock_m: float | None
    bearing_from_dock_deg: float | None
    eta_seconds: float | None
    recommended_altitude_m: float
    within_geofence: bool
```

A metre-accurate projection and a zone-centroid fallback are both useful. Confusing them puts a responder at the wrong end of a yard, so `source` and `accuracy_m` are not optional fields.

5.3 The rule language

```

id: loitering
severity: high
conditions:
  - kind: class_is
    labels: [person]
  - kind: zone_in
    zones: [main-gate, fence-line, substation, restricted-core]
  - kind: dwell
    seconds: 120
  - kind: time_between
    start_hour: 22
    end_hour: 5
cooldown_seconds: 300
visual_predicate: "a person standing still or pacing near a gate at night"

```

Twelve condition types, including genuinely temporal ones: **dwell**, **sequence** (A then B within T), **count_in_window**, **absence_of_person**, and **baseline_anomaly**. None of those can be expressed by a per-frame predicate, which is why the engine keeps per-entity state.

6 Failure modes designed for

Failure	Design response
Provider rate limit	Self-limit below the cap; a wait is cheaper than a 429 plus backoff
Provider cold start (83s measured)	Cross-provider failover; per-tier timeouts, not one global
Provider unavailable	Circuit breaker; degraded scene graph rather than a dropped frame
No API key at all	Cassette replay from committed fixtures — byte-identical output
Extension will not load	numpy fallback for vector search
Projection impossible	Return <i>no</i> coordinate; never fabricate one
Zone flicker at boundaries	Nesting awareness plus a two-observation debounce
Label specificity varies	Synonym and supertype matching (“sedan”, “vehicle”)
Model returns malformed JSON	Lenient parse, one repair attempt, then caption-only

7 Security boundaries

Two boundaries enforced by code, not convention

Permissions. Tools that change state or move an aircraft cannot be executed by the agent. Proposal and approval are separate code paths, and a source-level test asserts the agent’s own loop cannot pass the approval flag.

Credentials. Model-provider keys are server-side only and never serialised toward the browser. The map tile key is necessarily public — the browser fetches tiles directly — and carries a `NEXT_PUBLIC_` prefix that says so. A `gitleaks` pre-commit hook makes committing a credential fail loudly.